



Chapter Notes: Heat Transfer in Nature

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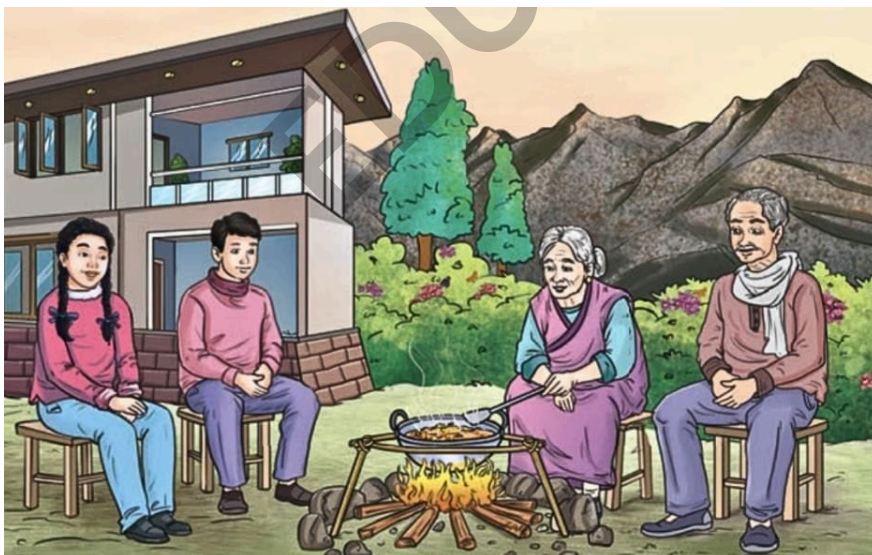
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Introduction

Pema and her brother Palden live in Gangtok. One cold winter evening, they sit near a fireplace. Palden talks about his trip to Kerala during winter and says Kerala is warmer and more humid than Gangtok.

Their grandfather explains, "Kerala is closer to the equator and has a long coastline, so it is warmer and more humid." Palden adds, "We learned that the Sun is the main source of heat and light, and places near the equator are usually hot."



While they talk, Pema watches her grandmother cooking *thukpa* in a big metal pan. She asks, "Why are cooking utensils usually made of metal?" Palden answers, "Because metals are good conductors of heat," which they studied in their science chapter.

Let us now study the nature of heat and how heat is transferred in nature.

Conduction of Heat

Conduction is the process by which heat transfers from the hotter part of an object to the colder part through direct contact. In solids, especially metals, particles vibrate when heated and pass this energy to neighbouring particles without moving their positions. This makes conduction an important process in many everyday activities such as cooking and metalwork.

When one end of a metal object is heated, the particles at that end gain energy and vibrate more. These vibrations pass to adjacent particles, transferring heat along the object. For example, when a metal pan is placed on a flame, heat travels from the flame through the pan so the whole pan becomes hot.

Activity: Why Certain Materials Conduct Heat Better

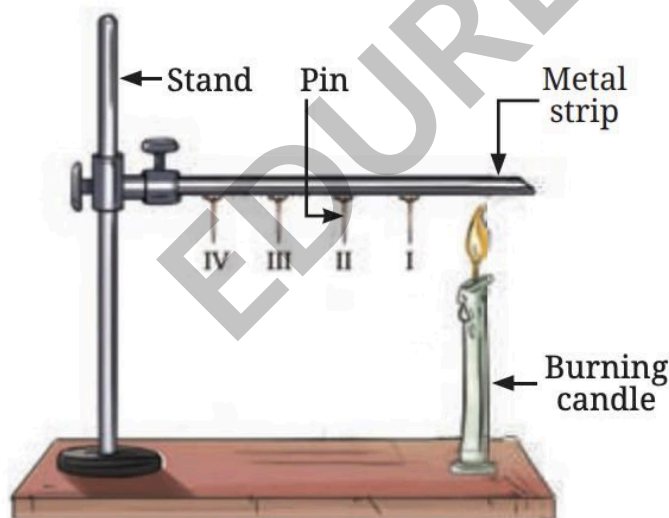
Materials Needed: Metal strip (aluminium or iron), four pins, candle or spirit lamp, stand (or two bricks), wax to attach the pins, heat source.

Steps:

Take a metal strip (about 15 cm long) and attach four pins to it using small drops of wax. Space the pins roughly 2 cm apart.

Secure the strip across a stand or between two bricks so the pins hang freely.

Heat the end of the strip that is away from the stand using a candle or spirit lamp. Observe which pin falls first, and note the order in which the pins fall.



Heat transfer in a metal strip

Prediction: Predict the order in which the pins will fall as the strip is heated.

Observation: The pin closest to the flame (Pin I) falls first, followed by the next pins in order (II, III, IV). The wax holding each pin melts in sequence because heat travels along the metal strip by conduction.

Conclusion

Conduction is the transfer of heat from the hotter part of a material to the cooler part through direct contact between particles.

In the activity, heat travels from the hot end of the metal strip towards the cooler end. The wax melts sequentially, causing the pins to fall one after another.

Metals are good conductors of heat, which is why metal utensils are commonly used for cooking.

Conductors and Insulators of Heat

Good conductors of heat:

Metals such as aluminium and iron allow heat to pass through them easily.

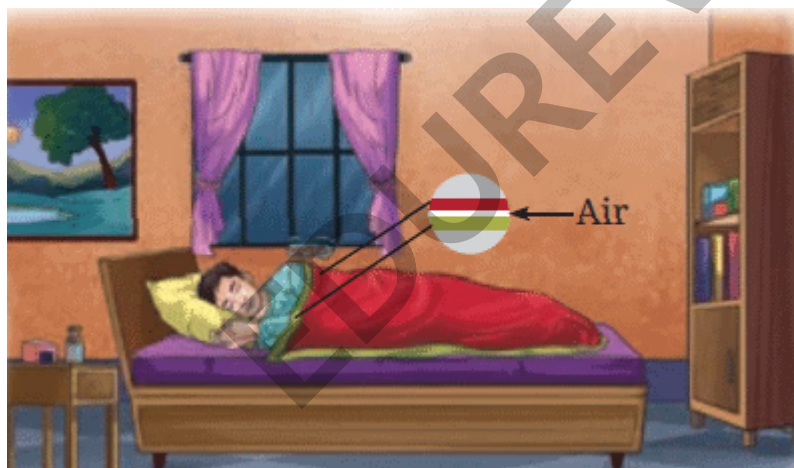
This property is why metal cooking utensils and pans are used to transfer heat efficiently from a flame to food.

Poor conductors of heat (Insulators):

Materials such as **wood**, **glass**, **clay**, and **porcelain** do not allow heat to pass through them easily.

Tea or coffee cups made of clay or porcelain retain heat longer, and woollen fabrics trap air to help keep us warm.

The presence of **air** between layers of clothing reduces heat flow and helps keep the body warm.



Air trapped between two thin blankets acts as an insulator

How Air Acts as an Insulator and Its Applications

1. Woollen fabric and heat retention

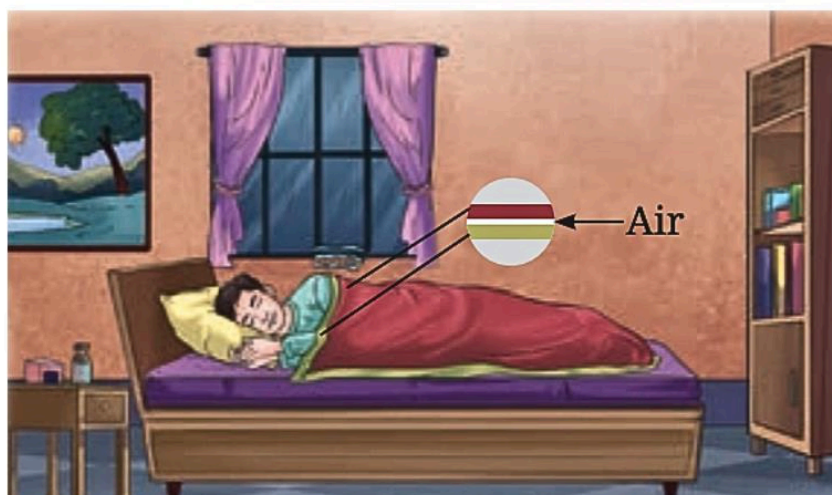
Woollen fabric traps air in tiny pores within the fibres.

Since **air is a poor conductor of heat**, it reduces the flow of heat from our bodies to the colder surroundings.

Trapped air acts as insulation; wearing several layers traps more air and keeps us warmer in cold weather.

2. Blankets and trapped air

The air between two thin blankets makes them warmer than a single thin blanket because the trapped air slows down heat loss from the body.



3. Insulated houses: Using heat transfer principles

Houses in very hot or very cold regions use insulation so indoor temperatures remain comfortable.

Materials and building methods are chosen to reduce undesired heat gain in summer and heat loss in winter.

4. Hollow bricks and heat insulation

Some houses use **hollow bricks** for outer walls.

Air trapped inside the hollow parts acts as a **poor conductor** of heat, reducing heat transfer through the wall.

This helps keep houses warmer in winter and cooler in summer by slowing the flow of heat.

Fascinating facts

The upper regions of the Himalayas, such as the **Mori block of Uttarkashi** in Uttarakhand, experience extremely cold climates and heavy snowfall in winter.

To keep houses warm in such conditions, people build walls with two wooden layers and fill the space between them with materials like **cow dung and mud**.

Both **wood** and **mud** are poor conductors of heat and act as natural insulation, preventing heat loss and keeping houses warm.



MULTIPLE CHOICE QUESTION

Try yourself: What is conduction?

- A Heat transfer through liquids
- B Heat transfer through direct contact
- C Heat transfer through air
- D Heat transfer through radiation

[View Solution](#)

Convection

Convection is the process of heat transfer in liquids and gases where heated particles move and carry heat with them. This movement forms currents that redistribute heat in the fluid.

Convection is a key process in many natural and everyday phenomena such as breezes near the sea, boiling water, and the rising of smoke.

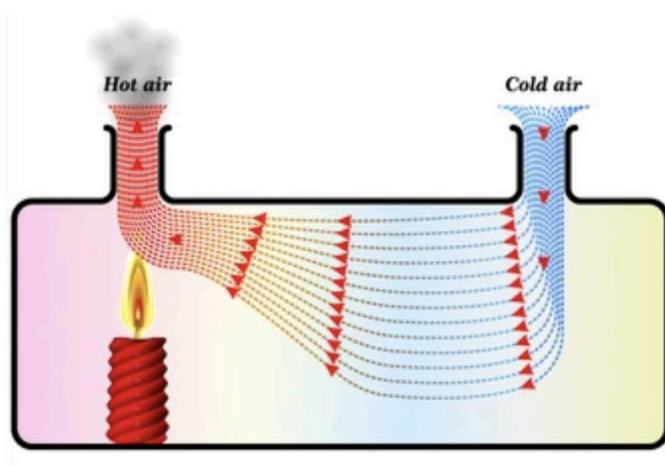
How convection works

When a liquid or gas is heated, the particles gain energy, become less dense, and rise. Cooler, denser particles then move in to take their place, creating a continuous cycle called a **convection current**.

For example, when water is heated in a pan, the warmer water rises while cooler water sinks, distributing heat throughout the pan.

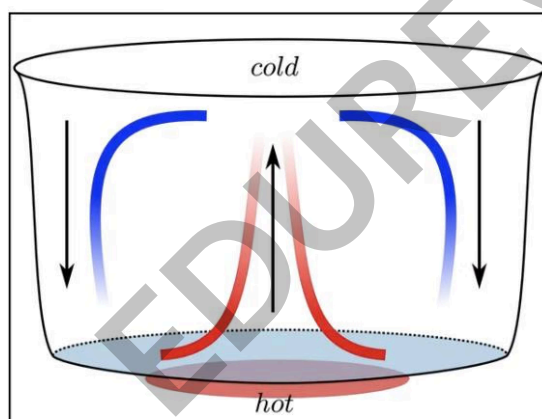
Convection in gases

In air, heated air expands, becomes lighter, and rises. Smoke rises because it contains hot gases and tiny solid particles released when something burns. These hot products are lighter than the surrounding air and therefore rise.



Convection in liquids

In liquids such as water, heated regions become lighter and rise, while cooler regions sink. This creates circular flow patterns that mix and distribute heat.



Activity: Why Smoke Rises (Paper Cup Experiment)

Materials needed: paper cups, a wooden stick, threads, a burning candle.

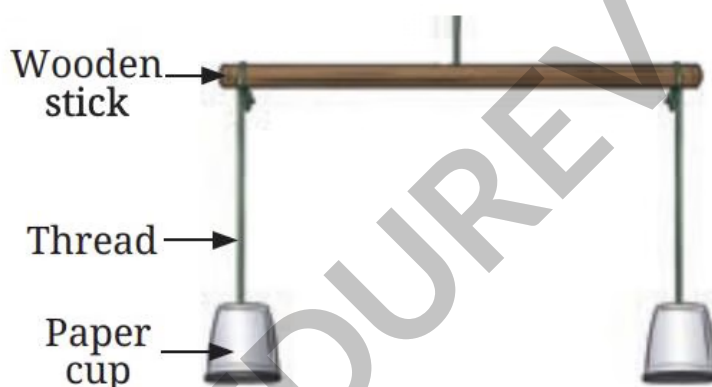
Procedure:

Hang the paper cups in an inverted position on the wooden stick using threads so they can move freely.

Place the stick horizontally so the cups can rotate or rise when air moves below them.

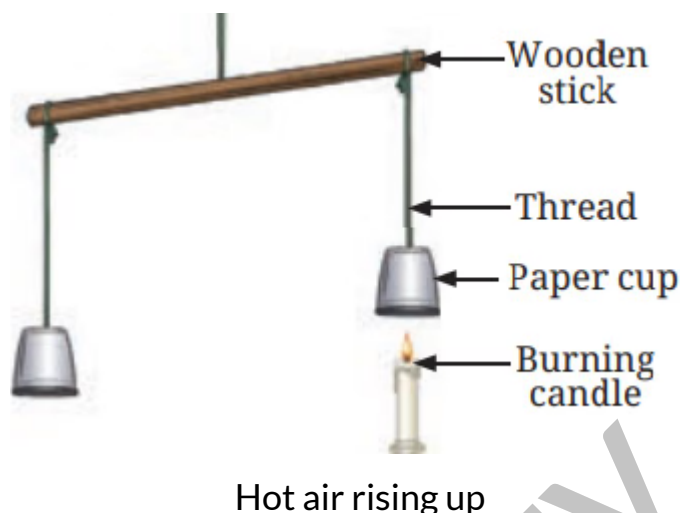
Light a candle and place it under one of the cups.

Observe the behaviour of the cups.

**Observations and explanation**

The cup with the candle under it rises because the air inside and below the cup is heated.

Heated air expands and becomes lighter, so the cup rises - a demonstration of how warm air moves upward in convection.

**Example of air expansion**

A partially inflated balloon placed in the sun expands because the air inside heats up and expands, making the balloon larger.

Smoke rising

When an incense stick burns, the smoke (hot gases and tiny solid particles) is warmer than the surrounding air and therefore rises by convection.

Activity: Convection in Water (Potassium Permanganate)

Materials needed: 500 mL beaker, water, a straw, a small grain of potassium permanganate (as a coloured tracer), candle.

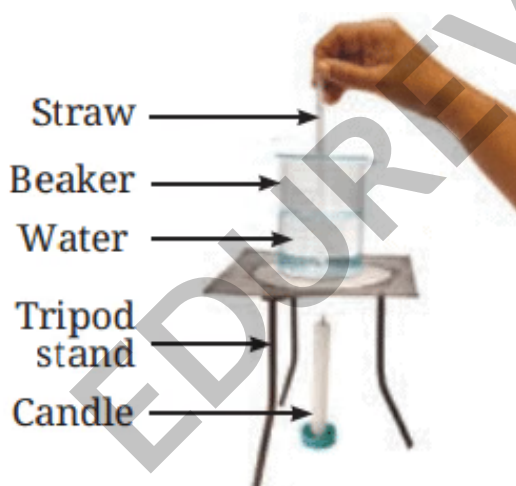
Procedure:

Fill the beaker halfway with water.

Use the straw to place a tiny grain of potassium permanganate at the centre of the beaker's base.

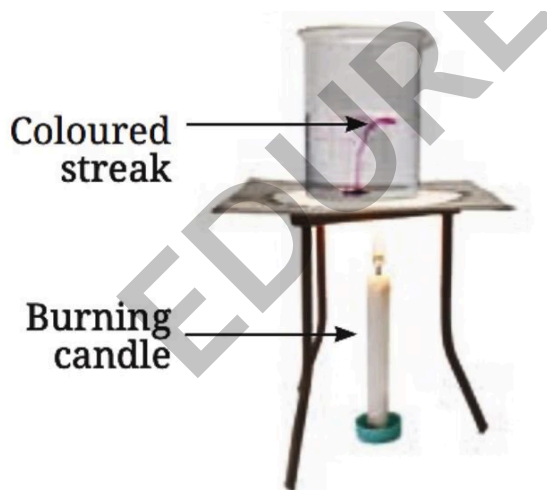
Place a candle under the centre of the beaker's base and gently heat.

Observe the movement of the coloured streak in the water.



Observations

A streak of colour moves up from the centre and comes down along the sides of the beaker as heating continues.



Demonstration of convection in heated water

Explanation and conclusion

The water at the bottom near the heat source becomes warmer, expands, becomes lighter, and rises.

Cooler and heavier water from the sides moves down to replace it, producing a circular motion.

This continuous motion is an example of **convection** in liquids and shows how heat is transferred by the bulk movement of fluid.

Land and Sea Breeze

During the day, land near the coast heats up faster than sea water because different materials heat at different rates.

Warm air over the land rises, creating a low-pressure area above the land.

Cooler, denser air from the sea moves in to fill this low-pressure area, producing a **sea breeze** felt on the coast during the day.

At night, land cools faster than the sea; cooler air over land flows out towards the warmer sea, producing a **land breeze**.

Activity: Comparing Heating of Land and Water

Materials required: Two identical bowls, soil, water, two laboratory thermometers.

Procedure:

1. On a sunny day, place soil in one bowl and water in the other, each to the same level.

2. Place a thermometer bulb in the soil and another in the water, ensuring they do not touch the sides or bottom.
3. Place both bowls in sunlight and observe the temperature rise over 20 minutes.

Observations

After 20 minutes, the soil's temperature rises more than the water's temperature - soil heats up faster than water.

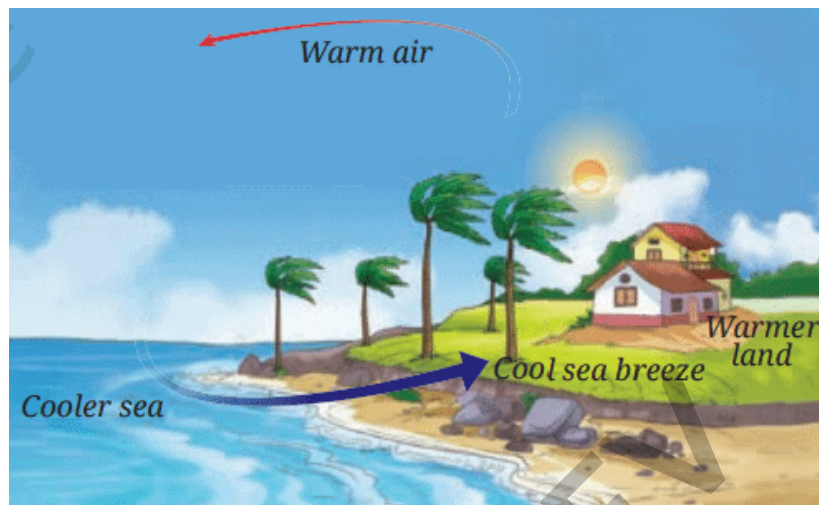
Cooling experiment

1. Bring the set-up indoors after heating and allow both to cool for 20 minutes.
2. Observe the rates of cooling.

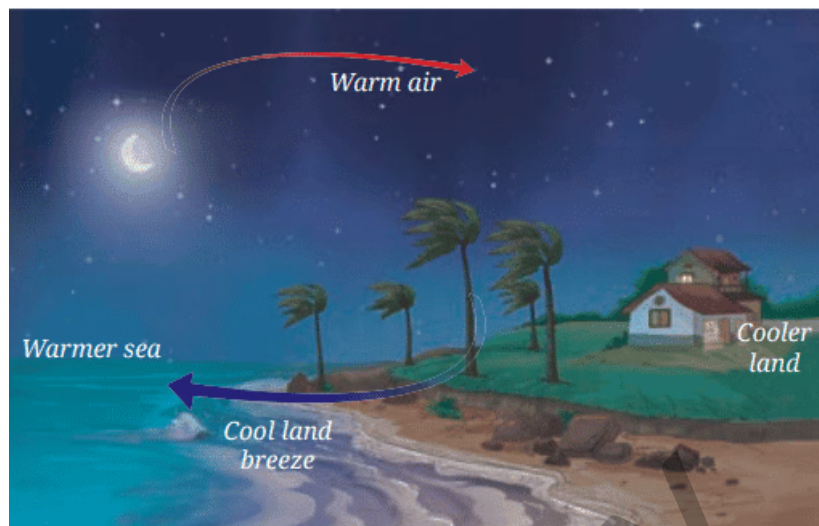
Conclusion

Soil heats up and cools down faster than water.

These differences cause land and sea breezes.



Sea Breeze



Land Breeze

MULTIPLE CHOICE QUESTION

Try yourself: What happens to air when it is heated?

- A It becomes less dense and rises.
- B It becomes denser and sinks.
- C It cools down immediately.
- D It remains the same.

[View Solution](#)

Radiation

Radiation is the transfer of heat through electromagnetic waves and does not require a medium.

Heat from a hot object reaches us by radiation directly; for example, the warmth we feel from a fire.

The Sun's heat reaches the Earth by radiation. The Sun's surface (about 6000°C) emits energy that travels through space as waves and warms the Earth.

Because radiation does not need a medium, we feel the Sun's warmth even though space is a vacuum.



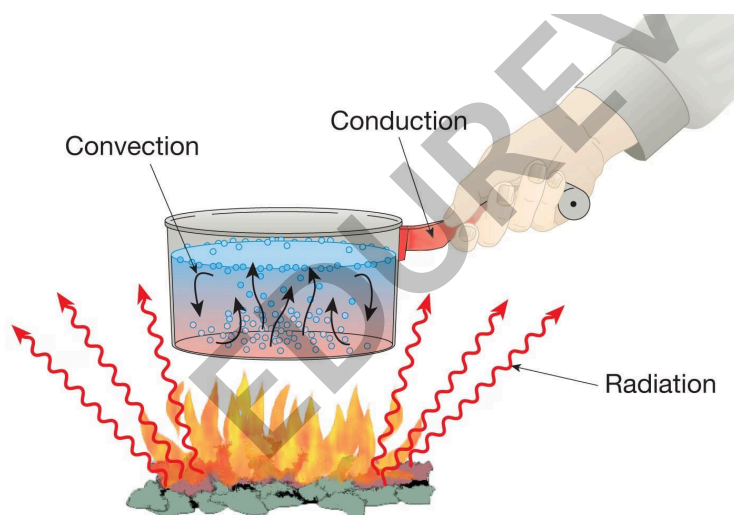
Examples of Heat Transfer in Daily Life (All Three Processes)

Many everyday situations involve conduction, convection, and radiation together. For example, when water is heated in a pan:

Heat moves from the flame to the pan by **conduction**.

Water inside the pan is heated and circulates by **convection**.

The warmth you feel around the flame and pan is partly due to **radiation**.



Fascinating fact: The Himalayan bukhari

In the upper Himalayan region, a traditional room heater called a **bukhari** is used to keep rooms warm in winter. A bukhari is an iron stove in which wood or charcoal is burned. A long pipe acts as a chimney to release smoke. The flat top of the bukhari can be used for cooking by placing utensils on it. When a bukhari is used, **conduction**, **convection**, and **radiation** all work together to warm the room and cook food.



Water Cycle

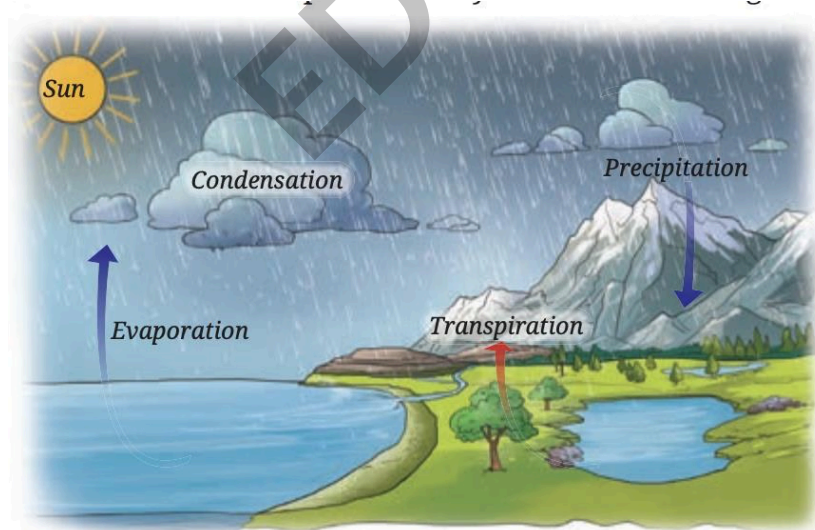
Water exists in three states in nature:

Liquid: in oceans, rivers and lakes.

Solid: as snow, glaciers and ice sheets in mountains and polar regions.

Gas: as water vapour in the atmosphere.

During summer, snow and ice melt due to the Sun's radiation and feed rivers that flow into oceans. Water from oceans, rivers and lakes evaporates due to the Sun's heat. Plants release water vapour through **transpiration**. Water vapour rises, cools, and condenses to form clouds. Clouds cause **precipitation**-rain, snow or hail. The continuous movement of water from the surface to the atmosphere and back is called the **water cycle**.



Water cycle

Importance of the water cycle

The water cycle redistributes and replenishes water on Earth. It ensures that water moves from oceans and other water bodies to the atmosphere and back again as precipitation. This cycle is essential for maintaining water supplies for ecosystems, agriculture and human use.

Know a scientist: Varahamihira

Varahamihira was a famous astronomer and mathematician of the 6th century CE from **Ujjaini** (now Ujjain), Madhya Pradesh. In his work **Brihatsamhita**, he described methods to predict seasonal rainfall based on observations of cloud formation, wind patterns, positions of stars and the moon, and other natural phenomena.

Seepage of Water Beneath the Earth's Surface

Activity: How Water Seeps Through the Ground

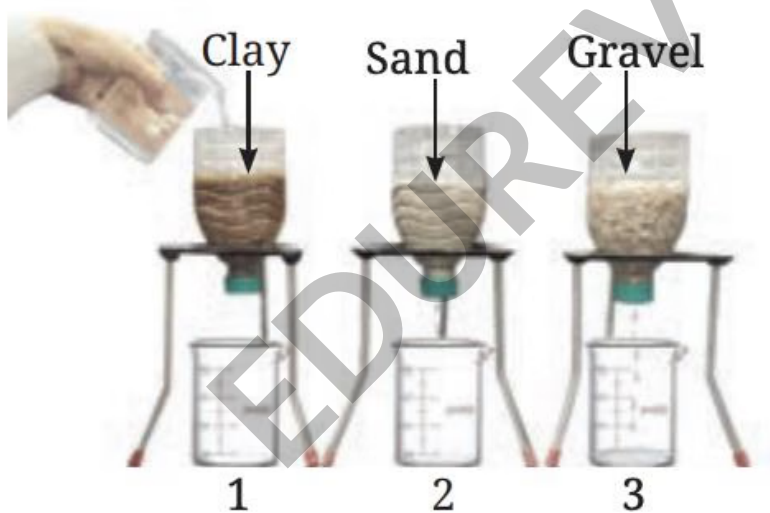
Take three transparent 1 L plastic bottles and cut them roughly in the middle.

Make a small hole in the cap of each bottle and keep them inverted.

Fill one bottle with clay, another with sand, and the third with gravel.

Place identical beakers below each bottle and add 200 mL of water to each bottle.

Collect and measure the water that flows out of each bottle for 10 minutes and compare the amounts.



An activity to compare the flow of water through
clay, sand and gravel

You will observe that water flows fastest through gravel, slower through sand, and slowest through clay. Differences in particle size and the spaces between particles explain this behaviour.

Gravel: Large spaces between particles allow fast flow.

Sand: Smaller spaces produce slower flow.

Clay: Very small spaces restrict water flow the most.

Theory and definitions

Infiltration is the process by which rainwater or surface water soaks down through soil and rocks beneath the Earth's surface.

The ease of infiltration depends on the size and connectivity of the pores (spaces) between soil and rock particles. Wider, well-connected pores allow faster seepage.

When water seeps into the ground it collects in the pores of sediments and in fractures of rocks; this stored water is called **groundwater**.

The underground layers of rock and sediment that store groundwater are called **aquifers**.

We access groundwater by digging wells or drilling bore wells into aquifers. Depth to groundwater varies from a few metres to hundreds of metres depending on location.

Groundwater is a vital resource but it is **not unlimited**; excessive extraction for agriculture and domestic use can lead to depletion.

Human impacts and conservation

Urbanisation reduces natural areas for infiltration: less vegetation and more concrete surfaces decrease the rate at which rainwater soaks into the ground.

As a result, groundwater recharge is reduced and groundwater levels may fall.

Methods to conserve and recharge groundwater include **rainwater harvesting** (collecting and storing rainwater) and building **recharge pits** to allow water to soak into the ground.

Water scarcity makes life hard, so people use methods to save and store water for dry periods.

In Ladakh, a practical solution called an **ice stupa** is used to store winter water as ice, which melts slowly in spring and supplies water for farming and daily use.

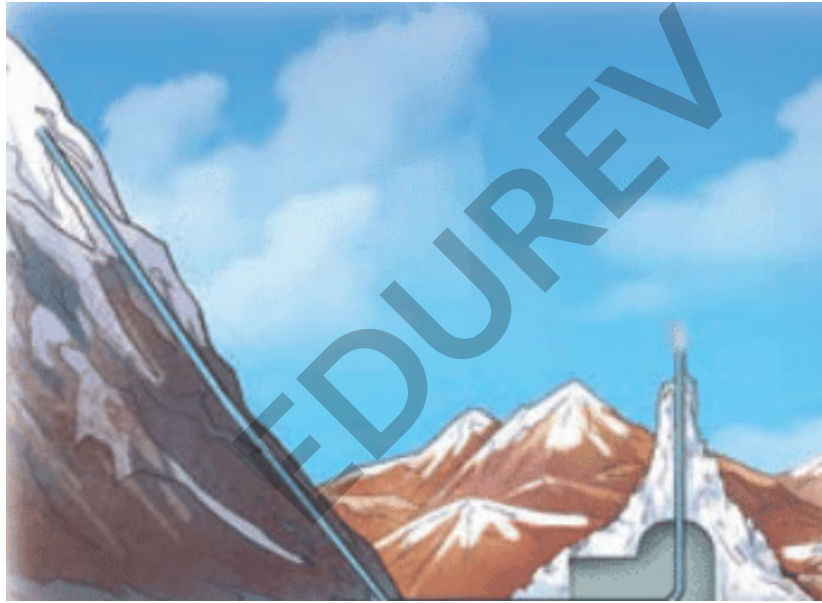
Science and Society: Ice Stupa

In Ladakh, spring streams often dry up because the sun's heat is not sufficient to melt mountain snow early in the year.

During winter, water from mountain streams is sent through pipes and sprayed into the very cold air.

This sprayed water freezes layer by layer to form a tall, cone-shaped ice structure called an **ice stupa**.

The ice stupa melts slowly in spring, releasing water gradually and thereby supporting farming and daily needs throughout the warmer months.



Ice Stupa

MULTIPLE CHOICE QUESTION

Try yourself: What process does the water cycle involve for returning water to the Earth's surface?

A Evaporation

B Infiltration

C Transpiration

D Condensation

View Solution

Terms to Remember

Conduction: Heat transfer through direct contact where particles pass energy to neighbouring particles without large-scale movement.

Convection: Heat transfer in liquids and gases by the movement of heated, less dense fluid that carries heat with it.

Radiation: Heat transfer through electromagnetic waves; it does not require a medium.

Insulator: A material (for example, wood, glass) that does not allow heat to pass through easily.

Evaporation: Process by which liquid water turns into vapour due to heat.

Condensation: Cooling of water vapour to form liquid droplets (cloud formation).

Precipitation: Release of water from clouds as rain, snow or hail.

Transpiration: Release of water vapour by plants.

Aquifer: Underground layer of rock or sediment that holds groundwater.

Seepage: Movement of water through soil or rock into the ground.

Bukhari: A traditional Himalayan heater using wood or charcoal for warmth and cooking.

Ice Stupa: A cone-shaped artificial ice structure used in Ladakh to store water in winter and release it in spring.

Summary

Heat is transferred in nature by three processes: **conduction**, **convection** and **radiation**.

Conduction occurs mainly in solids by particle vibrations and makes metals suitable for cooking utensils. Convection occurs in liquids and gases when heated, less dense regions rise and cooler material sinks, creating currents such as sea and land breezes. Radiation transfers heat by waves and does not require a medium; the Sun warms the Earth by radiation. The **water cycle**-evaporation, condensation, precipitation and seepage-is driven by the Sun's heat and redistributes Earth's water. Groundwater is replenished by infiltration into aquifers, but human activities can reduce recharge; therefore methods such as rainwater harvesting, recharge pits and innovations like ice stupas are important for water conservation.